Local vs global σ (Migkas $A/A_{\rm all}$, $\chi^2_{\rm null} = 45.14$)

Model	R^2	Local σ	Global σ
Point mass at CoE	0.521	4.4	4.1
Exponential, L free	0.759	5.3	5.1
Exponential, CatWISE 3% lock	0.716	5.4	5.1
Soft core $1/(1+(r/L)^{0.3})$, lock	0.743	5.4	5.1
Exponential (lock) + rotation & phase	0.845	5.5	5.3
Soft core + rotation & phase	0.870	5.4	5.1

Local: axis angle ℓ_0 taken as given. Global: ℓ_0 counted as one fitted parameter.